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Class : 12 Register Number

COMMON QUARTERLY EXAMINATION - 2025-26

BIOLOGY

Time Allowed : 3.00 Hours] [Max. Marks : 70

Part - I (Bio-Botany) (Marks : 35)

SECTION - I

I. Note. (i) Answer All the questions. 8×1=8

(ii) Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer.

1. Arrange the layers of anther wall from locus to periphery
   (a) Epidermis,middle layers, tapetum, endothecium (b) Tapetum, middle layers, epidermis, endothecium
   (c) Endothecium, epidermis, middle layers, tapetum (d) Tapetum, middle layers endothecium epidermis
2. Identify the correctly matched pair
   (a) Tuber - Allium cepa (b) Sucker – Pistia (c) Rhizome - Musa (d)Stolon - Zingiber
3. Extra nuclear inheritance is a consequence of presence of genes in
   (a) Mitrochondria and chloroplasts (b) Endoplasmic reticulum and mitrochondria
   (c) Ribosomes and chloroplast (d) Lysosomes and ribosomes
4. Select the period for Mendel's hybridization experiments
   (a) 1856 - 1863 (b) 1850 – 1870 (c) 1857 - 1869 (d)1870 - 1877
5. An allohexaploidy contains
   (a) Six different genomes (b) Six copies of three different genomes
   (c) Two copies of three different genomes (d) Six copies of one genome 6) In
6. In Which techniques Ethidium Bromide is used?
   (a) Southern Blotting techniques (b) Western Blotting techniques
   (c) Polymerase Chain Reaction (d) Agrose Gel Electroporosis
7. The Algae used in the production of single cell protein.
   (a) Yeast (b) Agaricus campestris (c) Cellulomonas (d)Chlorella
8. The time duration for sterilization process by using autoclave is ---------- minutes and the temperature is ----------
   (a) 10 to 30 minutes and 125° C (b) 15 to 30 minutes and 121° C
   (c) 15 to 20 minutes and 125° C (d) 10 to 20 minutes and 121° C

SECTION – II

II. Note : Answer any four questions. 4x2=8

9. 'The endosperm of angiosperm is different from gymnosperm'. Do you agree. Justify your answer.
10. What is the difference between missense and nonsense mutation?
11. Write short note on Atavism.
12. Name the chemicals used in gene transfer.
13. Define : Totipotency
14. Mention the types plant tissue culture based on the types of explant.

SECTION – III

III. Note : Answer any three questions. Q.No. 19 is compulsory. 3x3=9

15. Distinguish mound layering and air layering.
16. Name the seven contrasting traits of Mendel.
17. Write the salient features of Sutton and Boveri concept.
18. Give an account on Cryopreservation.
19. You are working in a biotechnology lab with a bacterium namely E.coli. How will you cut the nucleotide sequence? explain it.

SECTION – IV

IV. Note: Answer all the questions. 2x5=10

20. (a) What is endosperm. Explain the types. (OR)
    (b) Explain polygenic inheritance with an example.
21. (a) Mention the application of Biotechnology. (OR)
    (b) List out the applications of Plant Tissue Culture.

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PART – II (BIO-ZOOLOGY) (Max. Marks: 35)

SECTION – I

8x1=8

I. Choose the correct appropriate answer.

1. Which of the following set off diseases is caused by bacteria
   a) Typhoid and measles b) Typhoid and mumps
   c) Cholera and Tetanus d) Hepatitis and measles
2. A mRNA molecule is produced by
   a) replication b) Transcription c) Duplication d) Translation
3. Oral contraceptives are aimed at lowering the levels of which of the following?
   a) Progesterone b) LH c) FSH d) Both b and c
4. Human beings exhibit
   a) Hologamy b) Exogamy c) Isogamy d) Paedogamy
5. "Universal Donor" and "Universal recipient" Blood group are ——— and ——— respectively
   a) AB, O b) O, AB c) A, B d) B, A
6. Choose the correct order with respect to the events of human reproductive system.
   i) Cortical reaction ii) Capacitation iii) Spermiogenesis iv) Acrosomal reaction
   a) ii - i - iii - iv b) iii - ii - i - iv c) iii - ii - iv - i d) i - iv - iii - ii
7. The most common substrate used in distilleries for the production of ethanol
   a) Soyameal b) Groundgram c) Molasses d) Corn meal
8. Match the following and select the correct option.

Column I	Column II
A. Quarternary period	i) Age of dinosaurs
B. Devonian period	ii) Age of amphibians
C. Ordovician period	iii) Age of Fishes
D. Jurassic period	iv) Age of mammals

a) A - iv B - iii C - ii D - i
b) A - iv B - i C - ii D - iii
c) A - i B - ii C - ii D - iv
d) A - i B - iii C - iv D - ii

SECTION - II

II. Answer any Four questions. 4x2=8

9. Define regeneration mention its types.
10. List the major gases seem to be found in the primitive earth.
11. How will you detect the Foetal disorders during early stage of Pregnancy?
12. What are the Holandric Genes?
13. Differentiate: Leading strand and lagging strand.
14. Label the parts A, B, C and D for the following diagram



SECTION - III

III. Answer any Three of the following questions. Q.NO: 19 is Compulsory. 3x3=9

15. Describe the different phases of lifecycle in an organism.
16. What is Colostrum? Gives its significance.
17. Tabulate any three viral Sexually transmitted diseases with their causative agents.
18. Write short notes on Ideonella sakaiensis.
19. Who proved that human beings are superior to animals? In what way are they superior?

SECTION - IV

IV. Answer in Detail. 2x5=10

20. a) Give the schematic representation of Spermatogenesis and oogenesis in humans.

(OR)

b) Explain the Structure of the H_2L_2 molecule with a diagram.

21. a) List down the salient features of the genetic code.

(OR)

b) What will be the F1 and F2 generation for the following.
   i) Marriage between colour blind man and normal visioned woman.
   ii) Marriage between normal visioned man and colour blind woman.

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